MSE 433: Applications of MGTE

Module 3: Warehousing

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Mid-Semester Course Survey Results

- **KEEP**
  - Case-based format, open-ended nature, engaging and useful
  - Relating MGTE concepts to real scenarios
  - Providing consistent feedback and guidance in class
  - Providing detailed feedback on presentation and report grades
  - Interactive, engaging modules
  - Providing lots of time to ask questions in class, working sessions, in-class acitvities
  - Being passionate, caring so much about the course!
  - Easy mark! Flexible on deliverable formats
  - Being responsive and open to student feedback (accommodations)
  - Using bell! Walking around the class to quiet down noisy people
- **STOP**
  - Presentation file due Tuesday EOD is too tight.
  - New group formed for Modules 3-4. Allow at least 1 repeat.
  - Too many modules, workload, too many deliverables, deadlines close to each other.
  - Peer feedback. Just ask to bring major issues to the instructional team directly.
  - Presentation time too short. (10 sec – 1 min extra asked). Symposium instead?
- **START**
  - Fewer Modules, more time per module
  - More time for discovery before design
  - Lower expectations for depth and breadth
  - Structure around group accountability
  - 10 min break during Thursday classes
  - More transparency on due dates and deadlines

Module 3: Warehousing

Case Credit

Adapted from:

Warehousing Analytics Case
Dr. Fatma Gzara, MSE 433 W24-25

Review: Stages of Learning with Cases

- **Step 1: Data and Discovery**
  - Case presentation
  - Problem statement
  - Connection to MGTE courses
- **Step 2: Modeling and Validation**
  - Decide on methodologies and solutions
  - Test, validate, and connect results
- **Step 3: Decision and communication**
  - Distill findings, adjust if needed, and prescribe clear decisions and actions
  - Presentation and feedback

Module Outline (Tentative)

- **Tuesday Feb 24 (Today)**
  - Overview of Warehousing and Sortation
  - Case Presentation
- **Thursday Feb 26**
  - Demo of conveyor belt
  - Teamwork: Solution ideation
- **Next week (March 3-5)**
  - Teamwork: Testing and Validation
- **Tuesday March 10:** → files due EOD Wed.
  - Teamwork: Presentation and report
- **Thursday March 12:**
  - Case Presentations

Step 1: Data and Discovery

Case presentation
Problem statement
Connection to MGTE courses

Step 2: Modeling and Validation

Decide on methodologies and solutions
Test, validate, and connect results

Step 3: Decision and communication

Distill findings, adjust if needed, and prescribe clear decisions and actions
Presentation and feedback

Module 3: Warehousing

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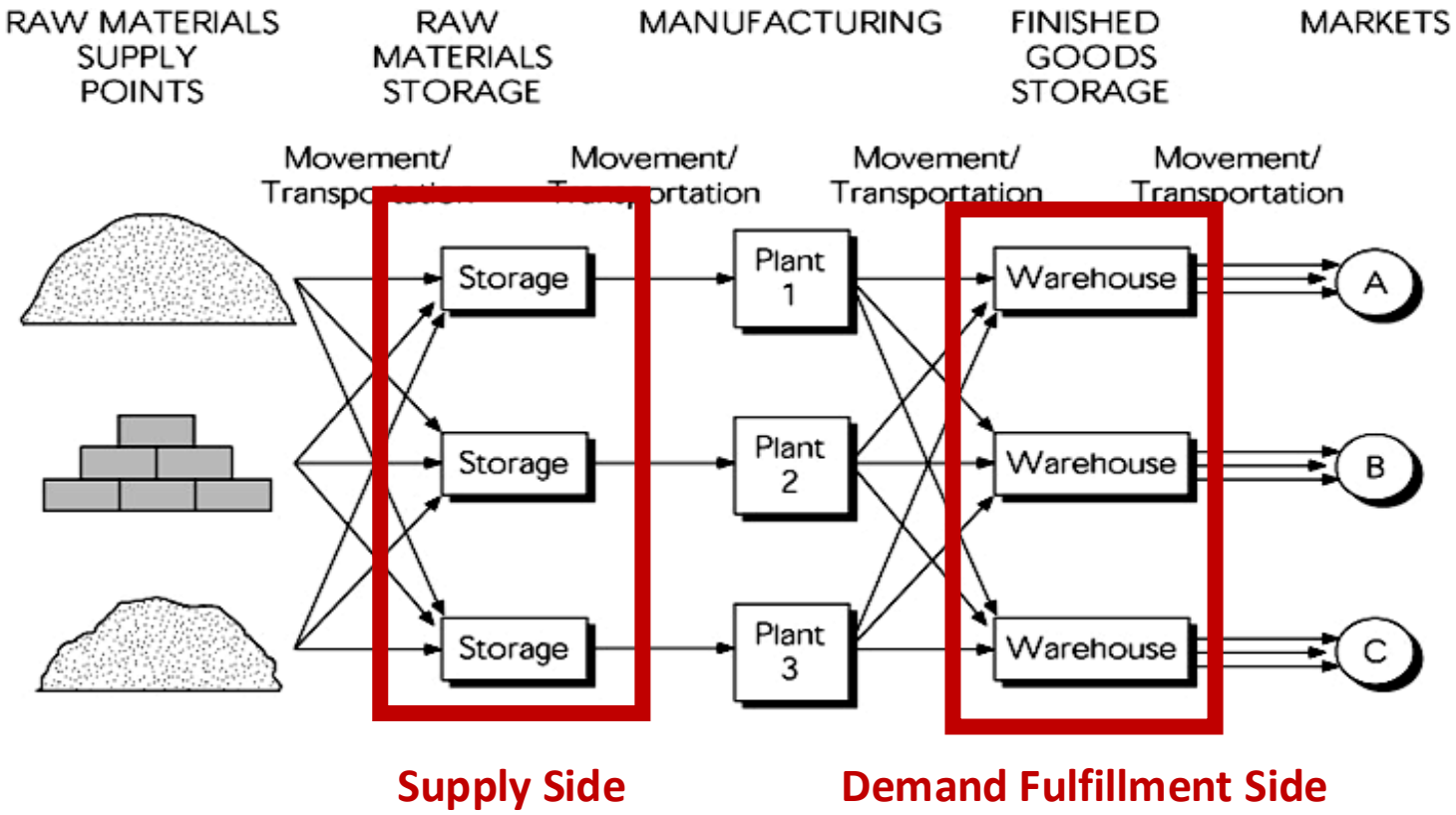
Introduction to Warehousing

- Warehousing is the intermediate storage and handling of goods in between two successive stages of a supply chain.



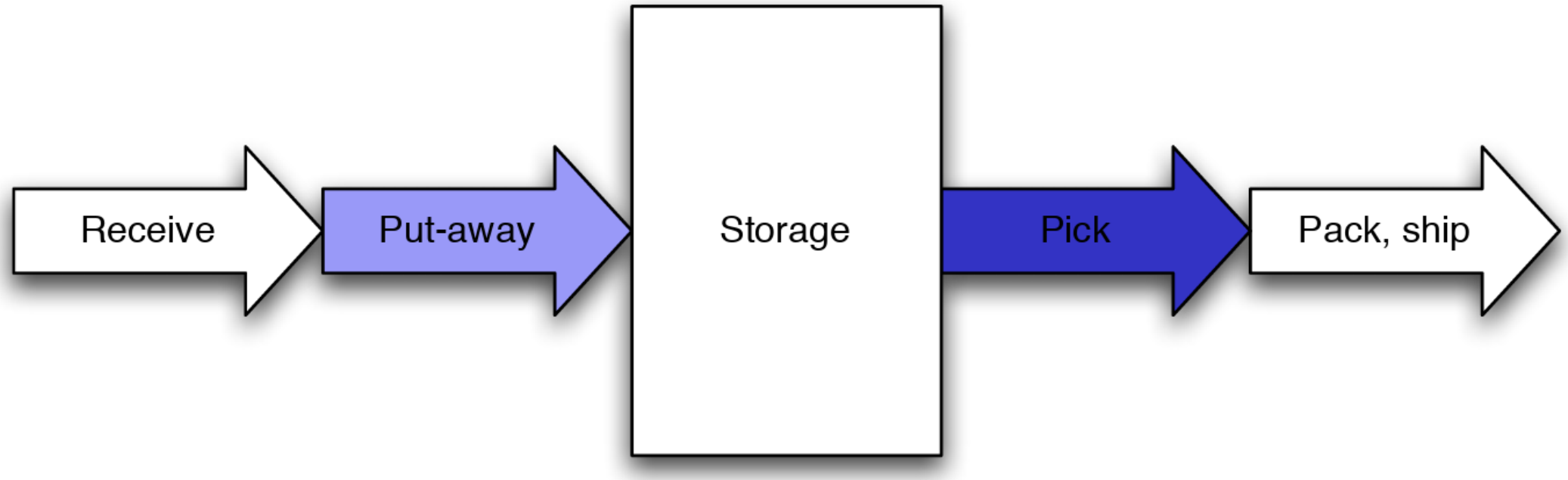
Warehousing Purpose

- **Supply side:**
  - Store, stage, and kit combinations of raw materials, work-in-process, and other components.

- **Demand Fulfillment side:**
  - Also called: **Distribution Center** or **Fulfillment Center**
  - Prepare and ship orders of products to end consumers, wholesalers, and/or distributors.

- We will be focusing on the **demand** side

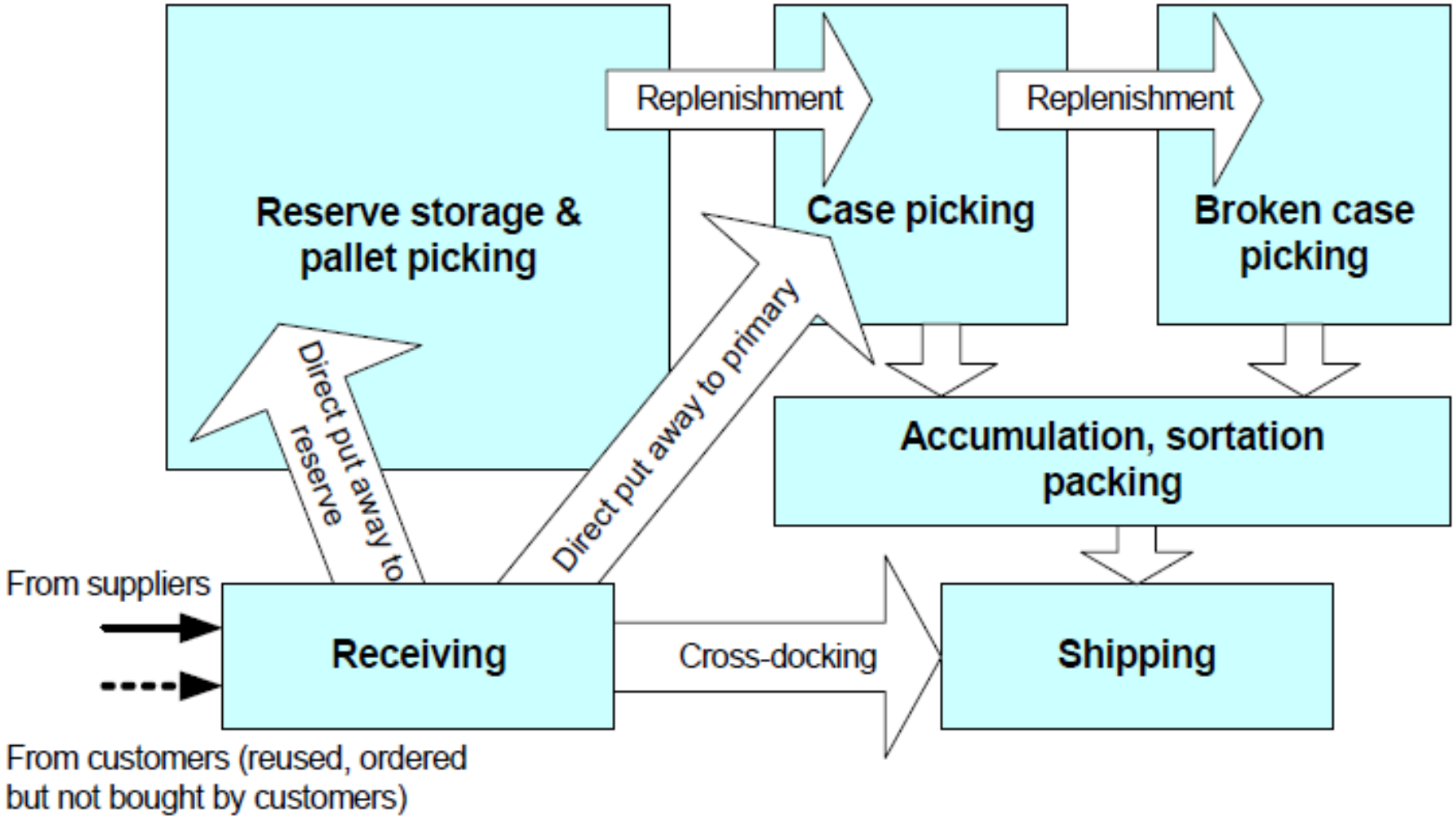
Main Warehousing Functions



- **Order Picking** is the most labor-intensive activity in most warehouses.
- Travel can be reduced by careful put-away.

Reference: Bartholdi III, J. J., & Hackman, S.T. (2008). *Warehouse and distribution science*. Supply Chain and Logistics Institute, School of Industrial and Systems Engineering, Georgia Institute of Technology.

Complex Function and Flows



Warehousing for E-Retailing

- **Small orders:** few order lines each demanding very few items.
  - At German Amazon warehouses, average order demand is 1.6 items
- **Large assortment:** much larger assortment, niche products
- **Tight delivery schedules:** Next-day or even same-day deliveries
- **Varying workloads:** scalable, dynamic capacity

- Increasing stress on warehouse operations
- Highly time-critical order fulfillment processes
- **Order Picking** is the most labor-intensive activity

Warehousing Methods

Mixed shelf storage

- Storage assignment strategy
  - Incoming loads are broken down into single units
  - Single units are spread all over the warehouse
- Applied by many large-size facilities
- Reduces distances from anywhere in the warehouse to the closest unit
- Multiple access points
- Completed orders are handed over to the central conveyor system
- Downsides:
  - Additional effort during the put-away of units
  - Larger order quantities of some SKU cause problems

Warehousing Methods

Batching and Zoning

To reduce the unproductive walking of pickers

- **Batching:** Multiple orders are unified into a batch of orders jointly assembled in a picker tour.
  - Need to be **separated** later.
- **Zoning:** The warehouse is partitioned into disjoint zones
  - Pickers pick the parts of an order stored in their zone
  - Requires **merging** of multiple partial orders later
- Used in:
  - **Order picking**: Trade-off between urgency and efficiency
  - **Intermediate storage**
  - **Consolidation area:** where orders are assembled for shipping

Warehousing Methods

Dynamic Order Processing

- Allow pick list updates after the picking tour has started
- Incoming urgent orders are instantly added to the current picker route
- Decide whether to add a new order to the current tour of a picker
- New orders may require altering the current picker tour
- Picked units have to be consolidated after picking into distinct orders

Warehousing Methods

AGV-assisted Picking

- Automated Guided Vehicles (AGVs)
- Reduce unproductive walking of pickers
- AGV autonomously accompanies pickers on their way through the warehouse
- AGV autonomously returns to the depot when an order is complete
- A new AGV is requested to meet the picker at the first storage position of the successive pick list
- Low investment for upgrading a traditional forklift to an AGV
- Need to be consolidated after picking into distinct orders

Warehousing Methods

Shelf-moving Robots

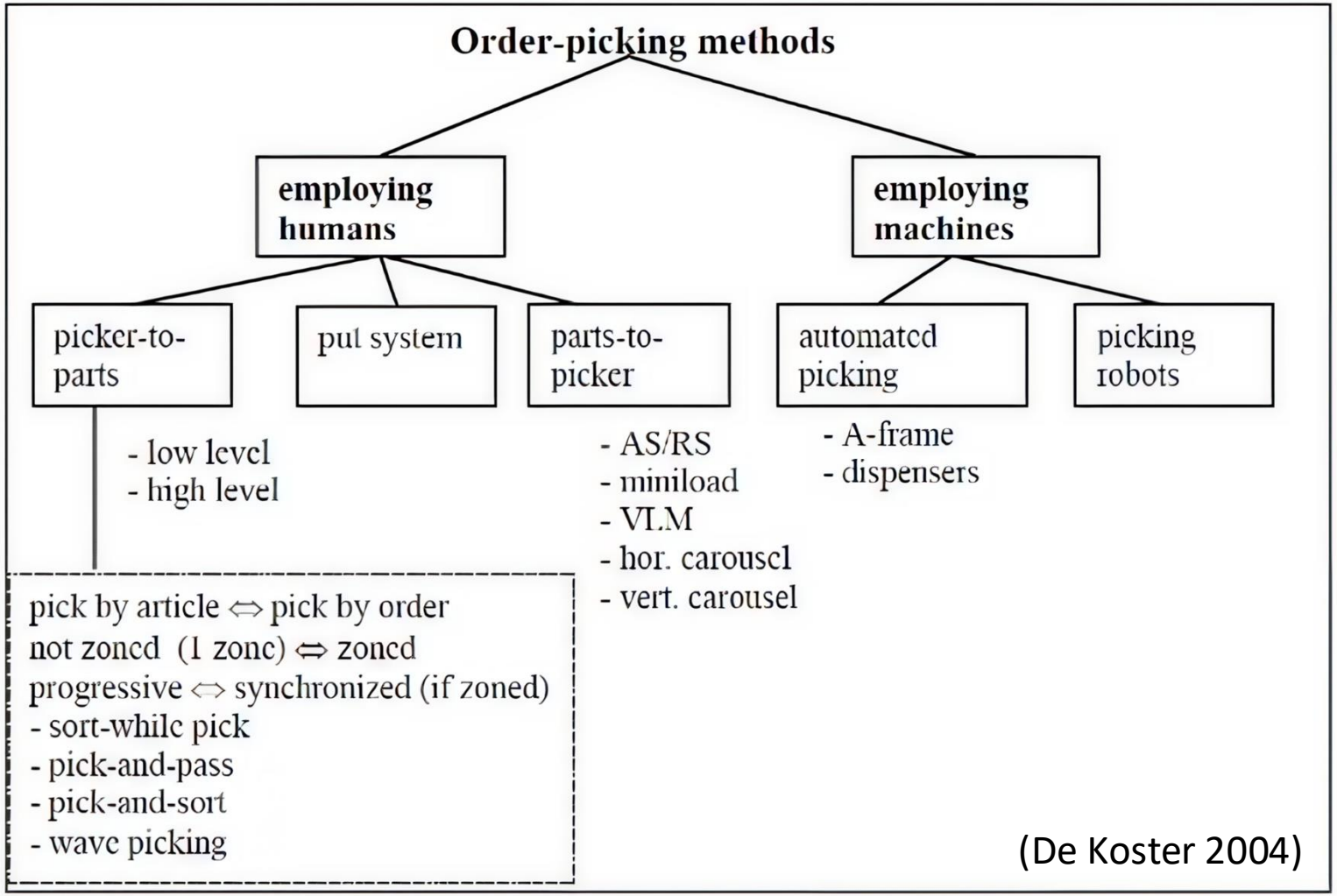
- Mobile robots lift movable racks (inventory pods) and bring them directly toward stationary pickers operating in workstations
- Shop floor is subdivided into a grid and each square is labeled with a barcode
- Multiple customer orders are processed concurrently
- Empty robots can drive unobstructed below the racks

Warehousing Methods

Advanced Picking Stations

- Designed and automatically fed with bins from a directly connected storage system via conveyors
- Order fulfillment system:
  - **Storage system:** Bins containing the SKUs are stored in an automated storage system
  - **Intermediate conveyor system:** Delivers requested bins between storage systems and stations
  - **Picking workstation:** Requested items are withdrawn by a human picker from the arriving storage bins and put into other bins that are associated with customer orders
- Downsides:
  - Requires plenty of constructional effort
  - Lacks scalability

Order Picking Systems

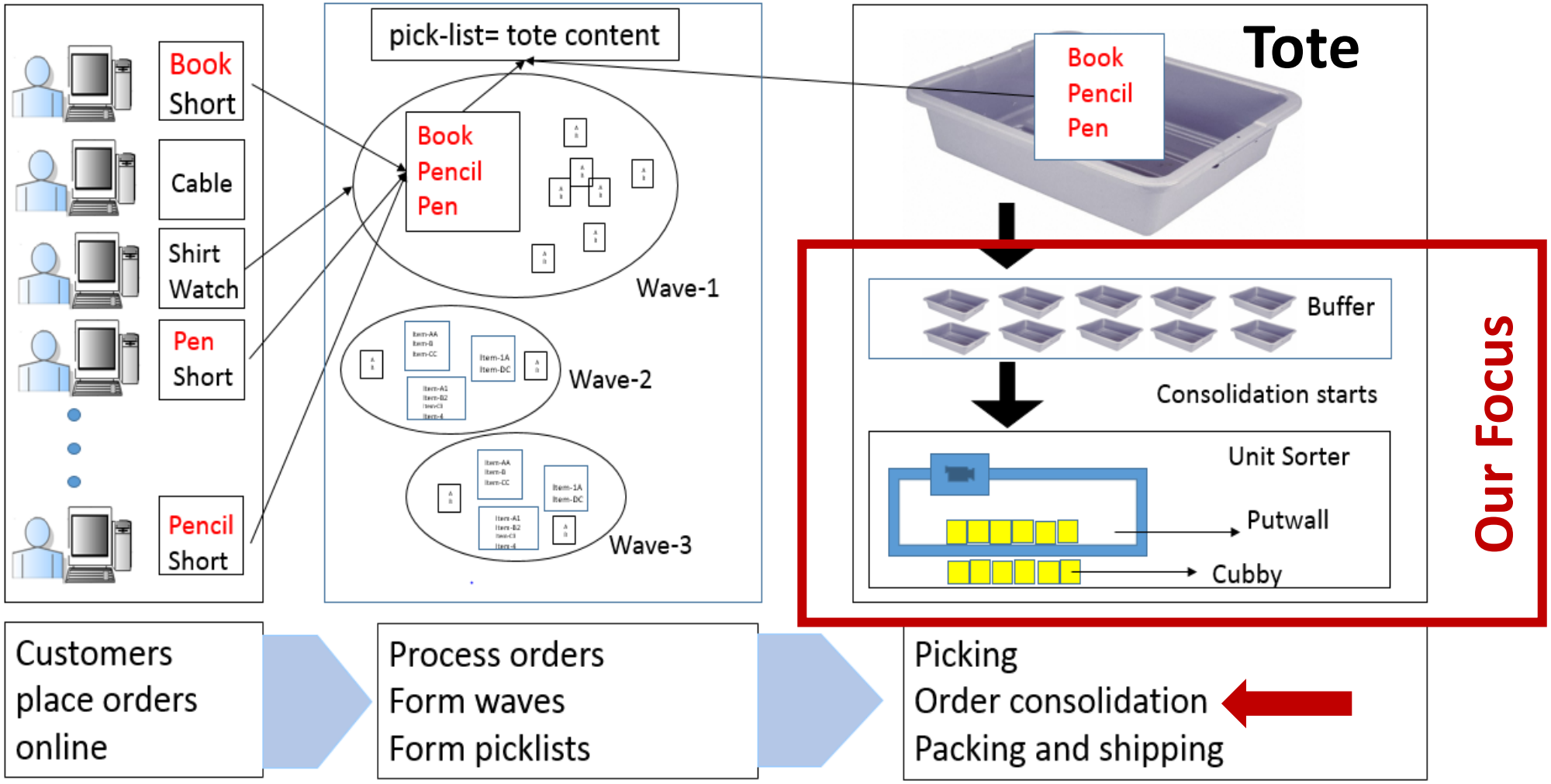


(De Koster 2004)

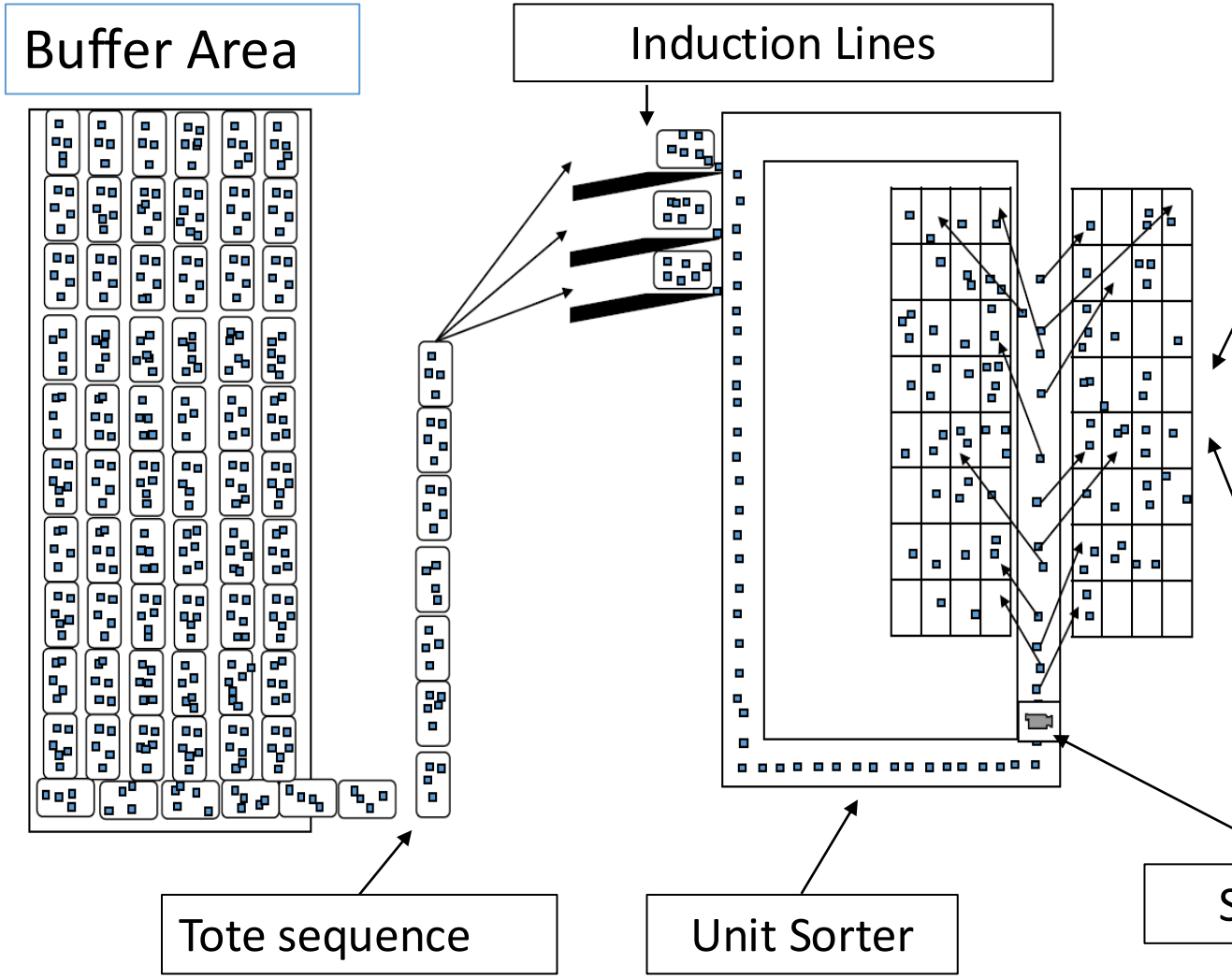
Objectives and Decisions

- Main goal: Maximize **service level**
  - Minimize order delivery time, Maximize order integrity and accuracy
- **Objectives:**
  - Minimize travel time/distance
  - Minimize throughput time for orders
  - Maximize the use of space/equipment/labor
- **Decisions**
  - Layout design
  - Storage assignment
  - Batching and zoning
  - Routing
  - Order accumulation/sorting

Typical E-warehouse Operation



Order Consolidation



Putwall , Pack side

Putwall, Put side

Tote Sequencing Problem

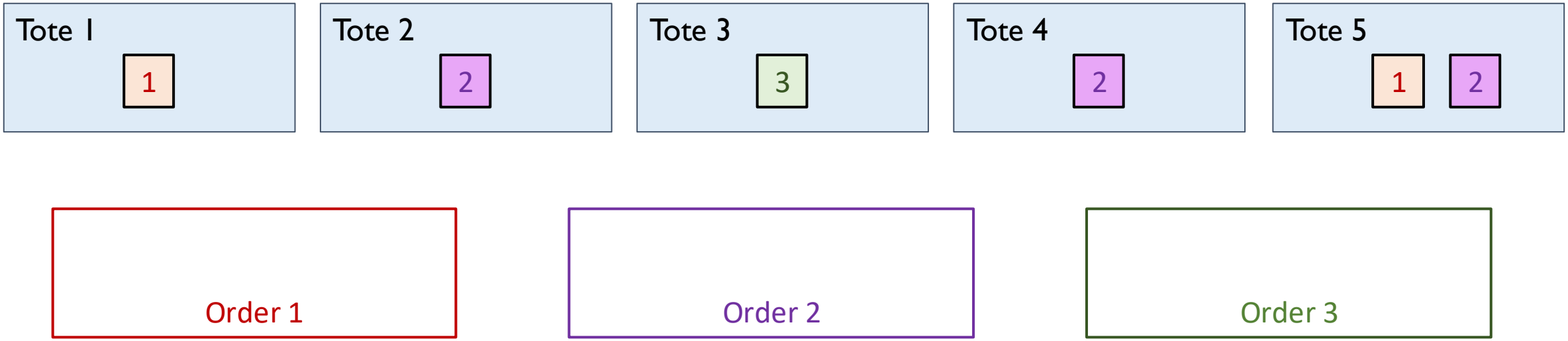
- In which order should the totes be processed?

- Objectives:
  - Minimize total tote completion time
  - Minimize total order completion time
  - Minimize maximum order completion time
  - Maximize early throughput (fastest first)
  - Minimize makespan
  - …

Tote Sequencing Problem - Example

- Consider 3 Orders to be consolidated from 5 Totes:

Tote (j)	1	2	3	4	5
Order number (k = 1, 2, 3)	1	2	3	2	1, 2
Tote j Processing time (p_j)	1	1	1	1	1



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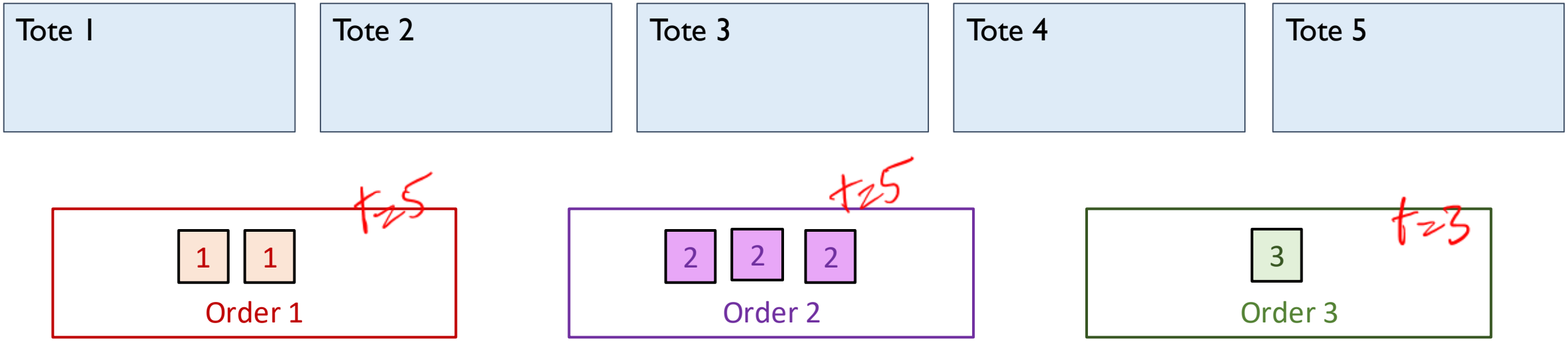

Tote Sequencing Problem - Example

- What if we changed the order?

Tote (j)	1	2	3	4	5
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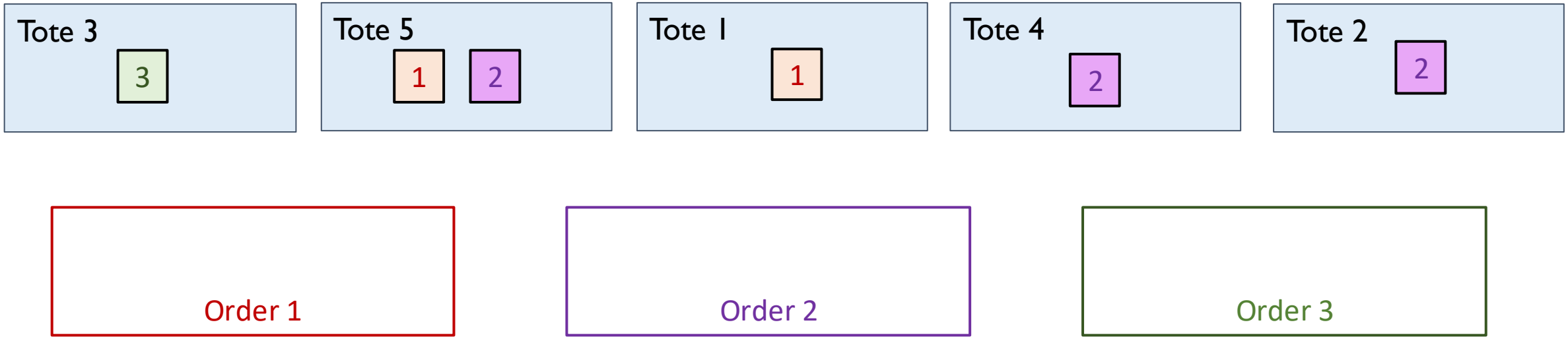

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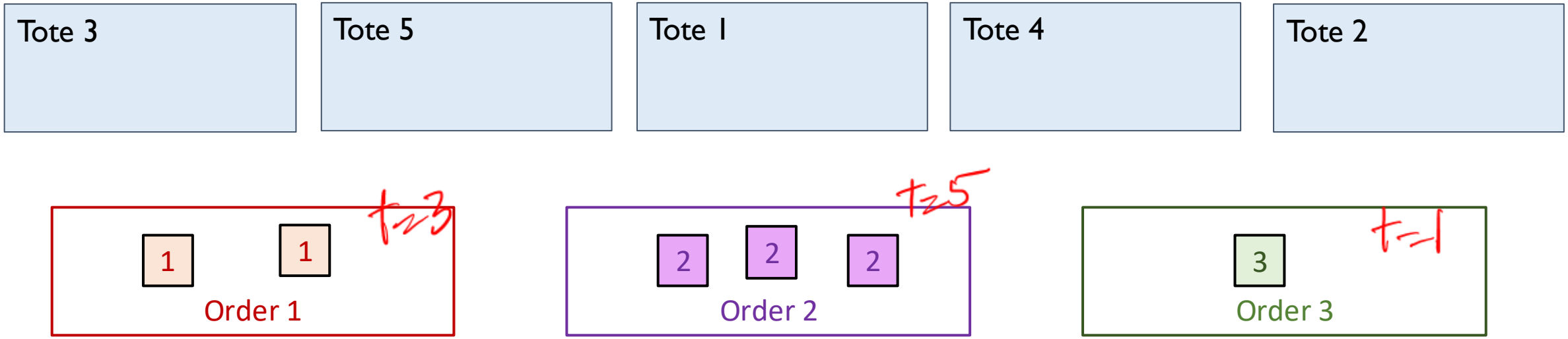
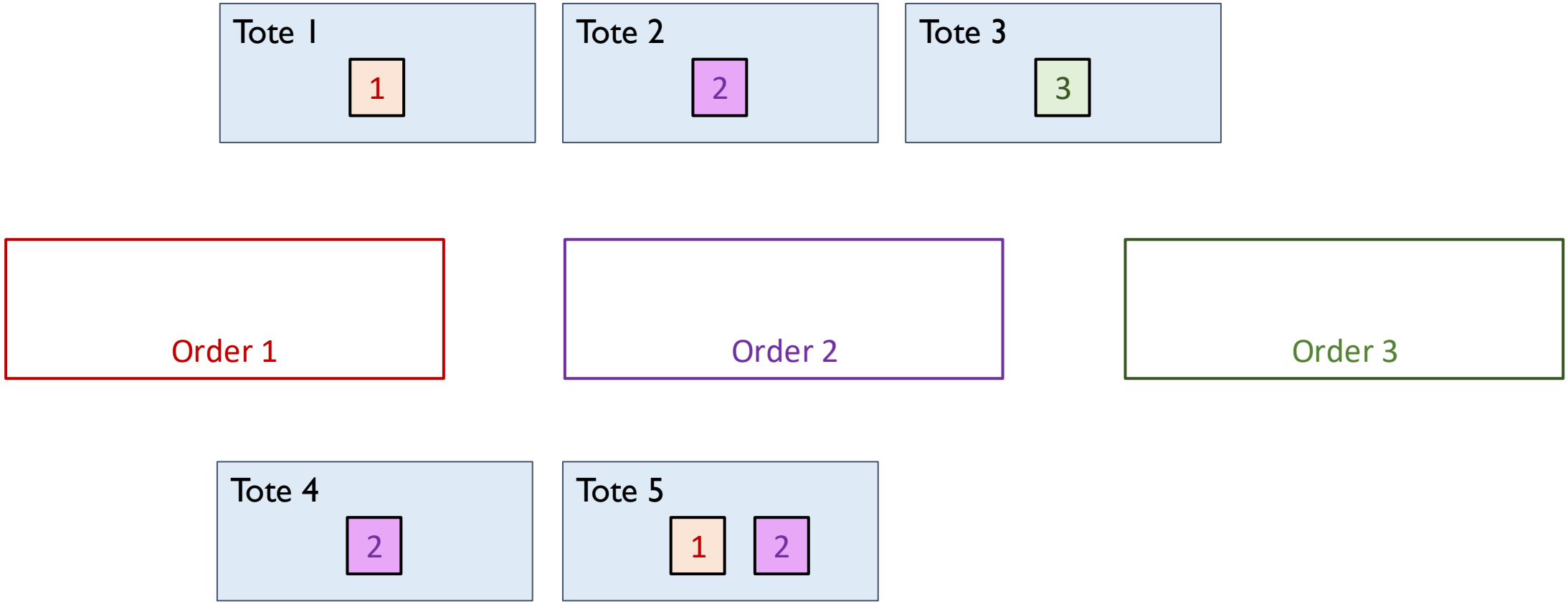


Tote Sequencing Problem - Example

- What if we had **two** induction lines?



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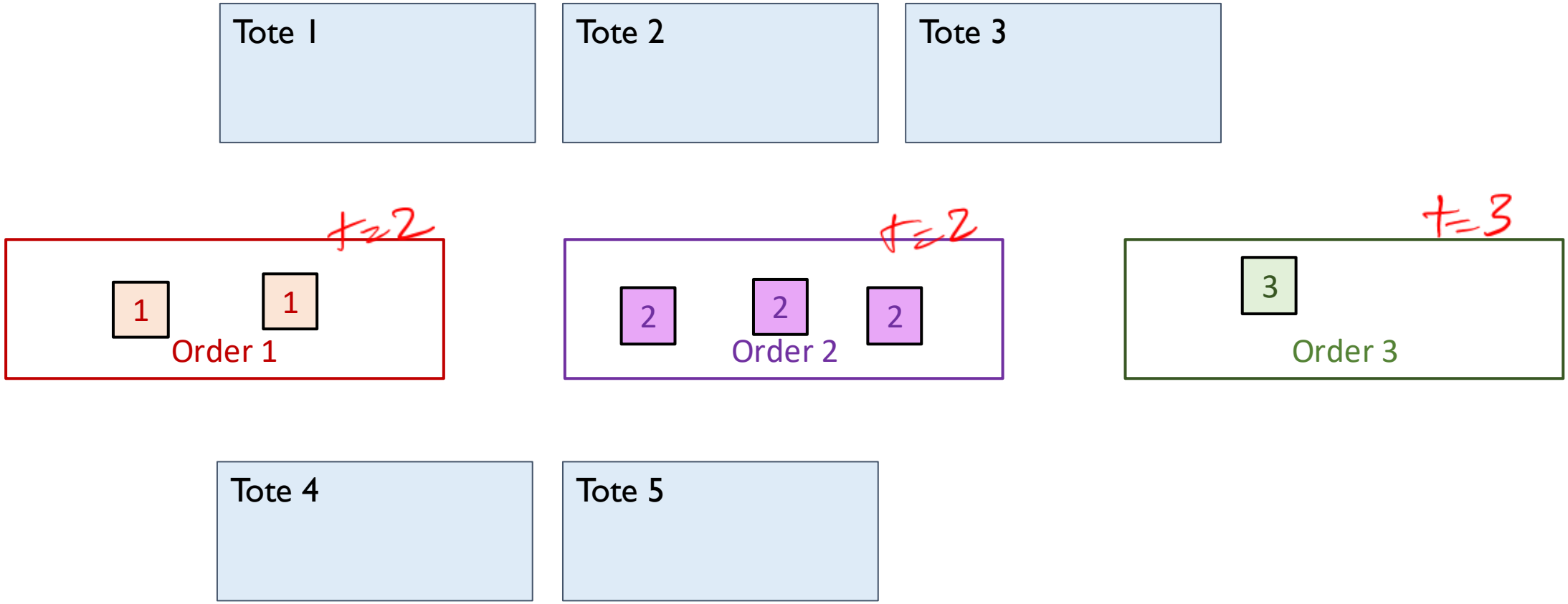
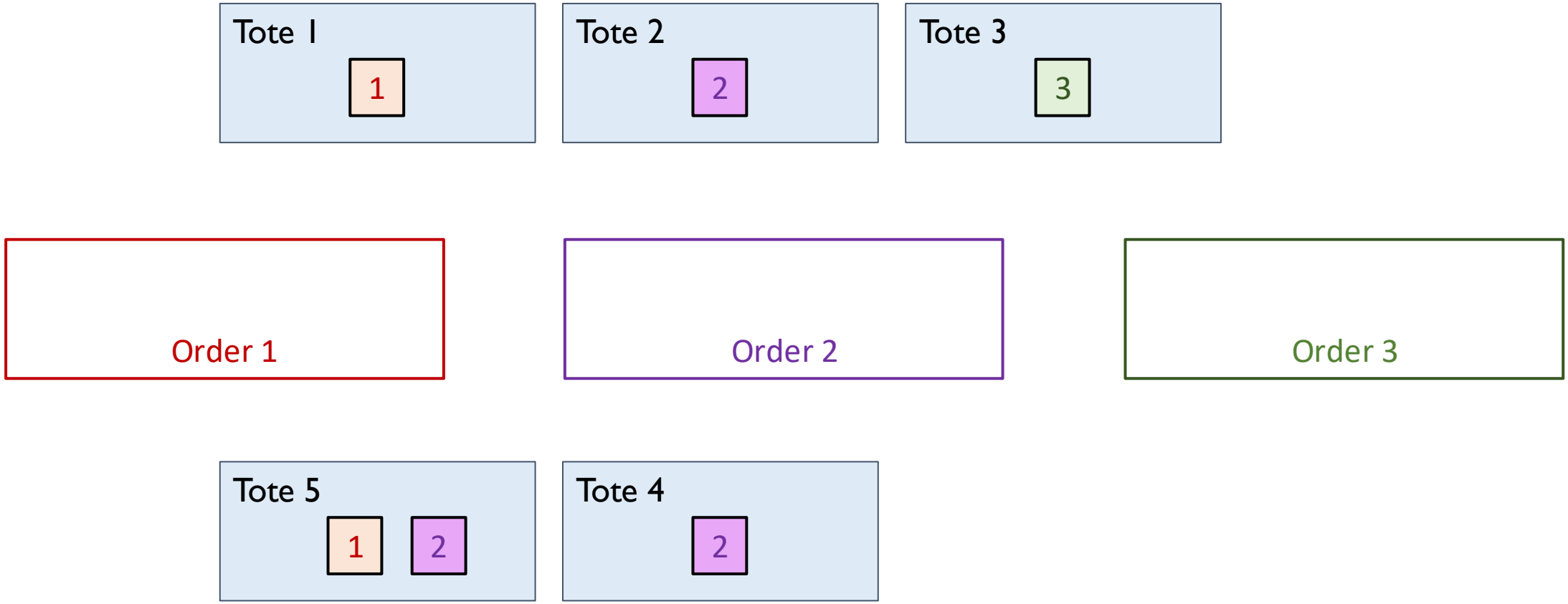


Tote Sequencing Problem - Example

- What if we had **two** induction lines? Let's switch Totes 4, 5!



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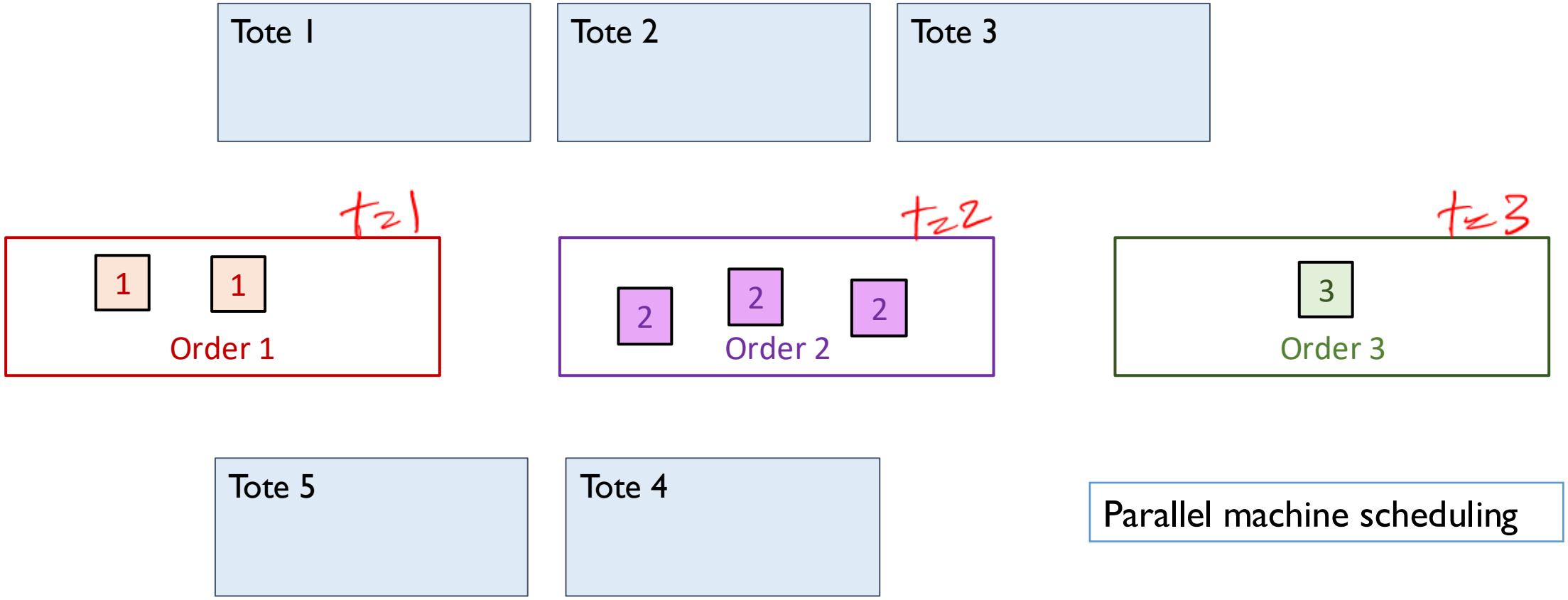


Parallel machine scheduling

Importance of Tote Sequencing

- It affects different objectives in different ways
  - Total tote completion time
  - Total order completion time
  - Maximum order completion time
  - Earliest completion time
  - Makespan
  - …

- What other objectives can you think of?

More Complex Cases

- Large number of item types
- Interchangeable items among orders
- Large number of totes
- Different processing times per tote and per item

Tote (j)	1	2	3	4	5
Order number (k = 1, 2, 3)	1	2	3	2	1, 2
Tote j Processing time (p_j)	3	1	2	4	5

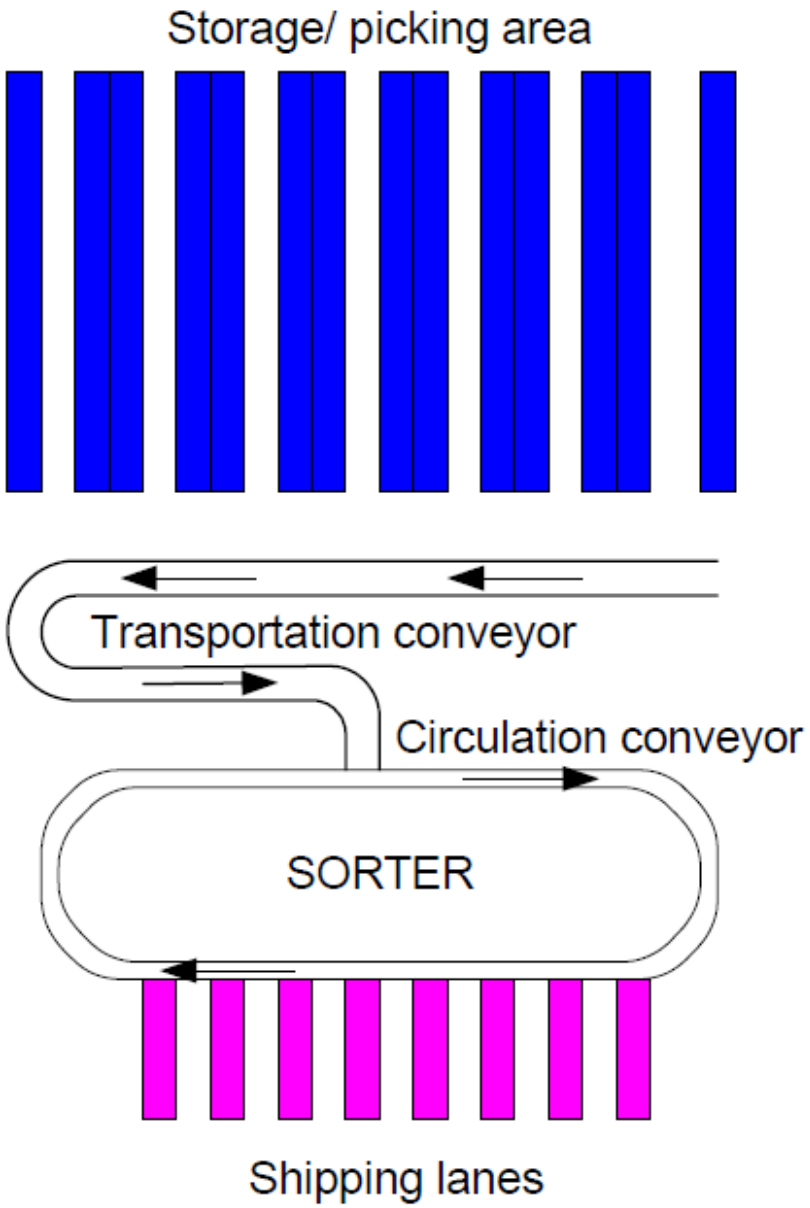
- Tote Sequence can be optimized!
- If more orders than fits put wall, orders need to be sequenced too.

Sortation Systems

- Split picked batches of items
- Consolidate the items per customer order
- A set number of lanes
- One order per lane at a time
- Scanners can be used to identify items

Decisions:

- Sequence of Totes
- How to empty the totes on the conveyors
- Which lane is for which order at what seque



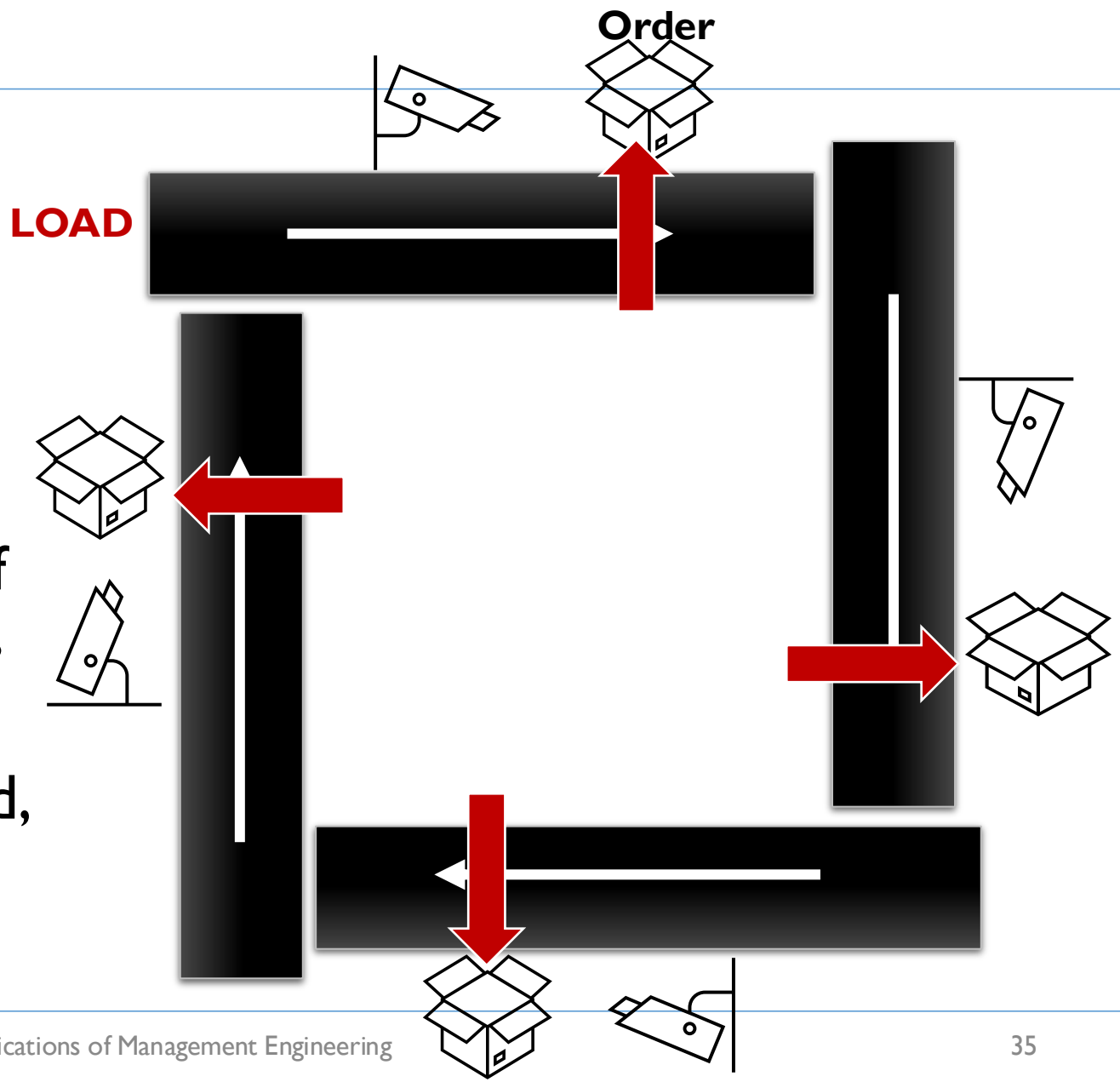
Module 3
Warehousing

IDEAS Clinic Conveyor Belt System

Efficient Order Consolidation and Sortation

Case Description

- Four conveyors form a circulation
- One scanner per conveyor
- One lane per conveyor
- Pneumatic arms push items off the conveyor into lanes if the scanner detects that it is needed in the order
- When an order is completed, the next order is up

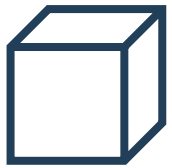


Case Description

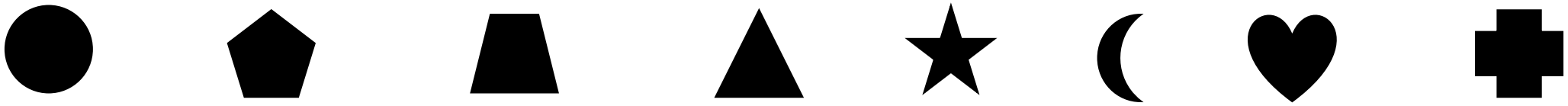
- Design the accumulation and sorting process to optimize performance indicators.
- You choose the performance indicator(s). Examples:
  - Total time to order completion
  - Circulation count
  - Errors in sortation
  - Throughput
  - …
- Use your MGTE skills to develop algorithms that operate the conveyors efficiently

Case Data

- A random data generator will be provided to create a set of orders and a set of totes that include all picked items for the orders.

- The items: 
- Differentiating "readable" icon on each item:

circle pentagon trapezoid triangle star moon heart cross



Case Data

- A reference data generator is provided, which creates 3 CSV files:
  - The types of items in each order
  - The number of items of each type in each order
  - The tote number where each item type per order is located
- An input template for you to modify, which controls the conveyor the IDEAs Clinic conveyor.
  - Template CSV file will be shared.
  - You MUST adhere to this template to run the conveyor belt simulations.
  - Demo on Thursday Feb 26.

Your Role

- Design your solution for the efficient sortation/consolidation operation for the IDEAS Clinic conveyor.
- Apply your design solution to randomly generated instances and calculate important statistics and performance indicators.
- Discuss the quality of your designs.
- Test your solution on the conveyor, report results, and assess/adjust your designs accordingly.
  - Use smaller instances for testing
  - Each team will have an opportunity to test their solution on the conveyor.
  - Sign-up sheets will be posted for next week (March 3-6)

Phase 1
Data and Discovery

Data

- Go to: LEARN > Content > Module 3: Warehousing > Data

For use in your design:

- Random data generator

For conveyor belt simulation:

- Example input file
- Example output file

Teamwork Time

The instructor and TAs are available to help or answer any questions!